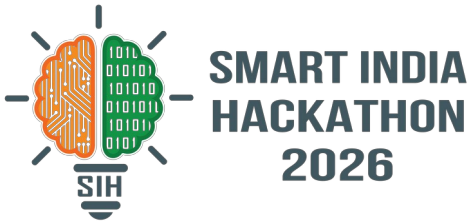


SMART INDIA HACKATHON 2026



RECORD / SIH26002 / TEAM-17

RASTA AI

Route Intelligence for Essential-Supply Logistics in North Eastern India

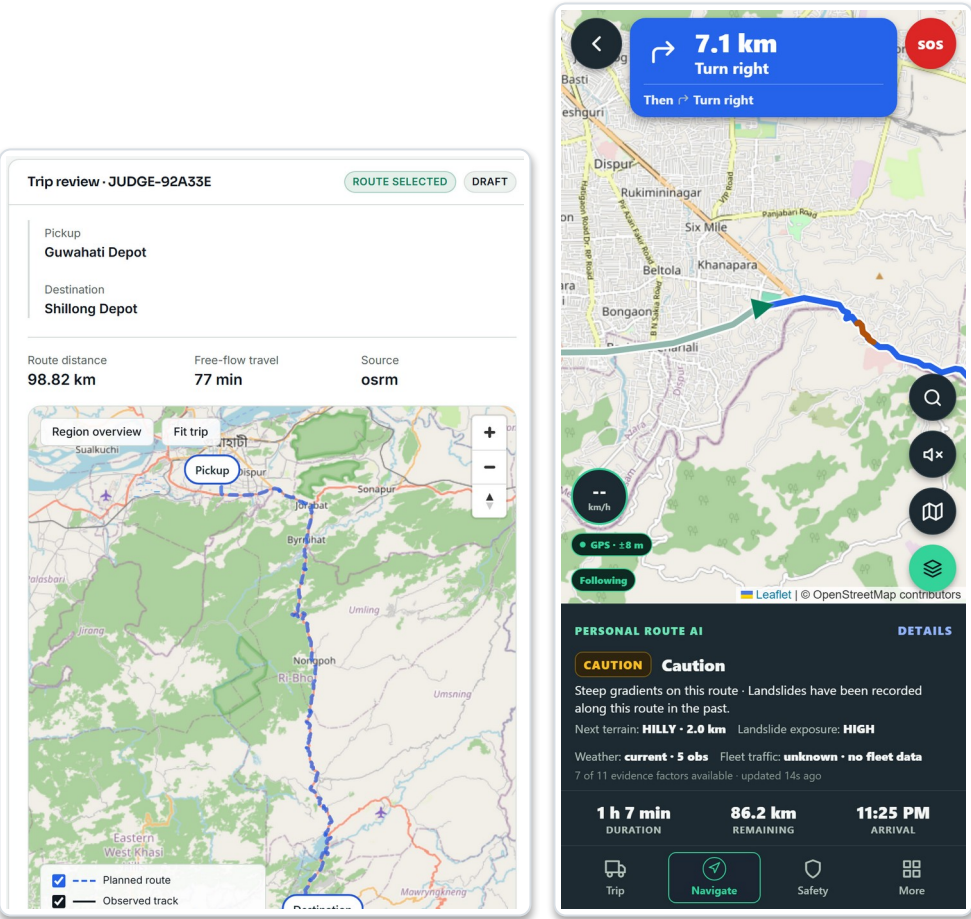
PS . ID	SIH26002
PS . TITLE	AI-Based Smart Logistics and Accessibility Intelligence Platform for North Eastern Region (NER)
THEME	Smart Automation
CATEGORY	Software
TEAM . ID	17 (internal group no.)
TEAM . NAME	NER-AI LOGISTICS
MINISTRY	MDoNER

SHORTEST IS NOT ALWAYS USABLE.

“Can this essential-supply truck reliably use this corridor now?”

STATUS: WORKING PROTOTYPE / PHYSICAL-PHONE VALIDATED / CORRIDOR GUWAHATI-SHILLONG

EXHIBIT A – LIVE WORKING SYSTEM



MANAGER ROUTE REVIEW / DRIVER NAVIGATION – REAL SCREENS

IDEA / PROPOSED SOLUTION

Not the shortest road. The road that is operationally usable now.

PROBLEM NER corridors close without notice: monsoon rain, landslides, floods, official warnings, single-lane hill roads, no phone signal. A shortest-path router still sends the truck.

BASELINE / NORMAL ROUTER

Origin → Shortest / fastest road → Truck

No terrain, weather, hazard or freshness check. No governance. No answer when data is missing.

RASTA AI / EVIDENCE BEFORE THE ROAD

Origin → real OSRM route, alternatives, turn steps → route-specific evidence

Terrain

Weather

Landslide exposure

Flood context

Official warnings

Fleet traffic

Evidence freshness

CONTINUE

CAUTION

HOLD

REROUTE

Manager governance → driver navigation, danger context, 23 languages, offline package

UNKNOWN ≠ SAFE Missing or stale evidence is marked UNKNOWN, never counted as safe, and the route goes to an authorised reviewer.

EXHIBIT B — CHECK CONDITIONS

REQUIRES REVIEW

Hazard evidence is incomplete or elevated. An authorised reviewer must review this route before selection. Refresh conditions after review.

Landslide data is not available - no source is connected. · Steep gradients on this route · Landslides have been recorded along this route in the past. · The landslide inventory ends in 2017 - recent years are not covered. · River levels near this route are close to their recent average. · No official disaster alert names a district on this route right now. · No fleet truck has driven this road in the last fifteen minutes, so traffic is unknown.

Unavailable: Fuel range, Landslide, Road quality, Truck restrictions

Evidence coverage · Weather AVAILABLE · Terrain AVAILABLE · Warnings AVAILABLE · Traffic UNKNOWN

TERRAIN0.5 KM STEEP

49–1436 m · 1975 m climb · steepest 10.3%

flat 61.8 km · rolling 29.6 km · hilly 7.0 km · steep 0.5 km

Copernicus DEM GLO-90 via Open-Meteo + SRTM 30 m via OpenTopoData · 100% of 198 samples answered

LANDSLIDE HISTORYHIGH EXPOSURE

22 recorded landslides within 5 km of the road, nearest 0.4 km

6 more nearby placed too imprecisely to count · HIGH is three or more; MODERATE is one or two

NASA Global Landslide Catalog · inventory 2007–2017 — aged, recent years not covered

RIVER LEVELSNORMAL

Discharge at 5 river cells along the corridor, highest 1.0× its own 30-day mean · GloFAS via Open-Meteo, 2026-09-14. A level, not a flood forecast and not a road-closure claim.

OFFICIAL ALERTSNONE ON CORRIDOR

NDMA SACHET CAP feed, 99 alerts nationwide at 10:13:54 PM · placed by district name (Dispur, East Khasi Hills, Kamrup Metropolitan, Myllem, Ri-Bhoi, Umling, Umsning) · 4 more active elsewhere in the corridor states, not placeable on this road.

FLEET TRAFFICUNKNOWN

REAL SCREEN / EVIDENCE / FRESHNESS / UNKNOWN

PIPELINE / VERIFIED DATA → DECISION → DRIVER

WEATHER	Open-Meteo · MET Norway
TERRAIN	Copernicus DEM · OpenTopoData
WARNINGS	NDMA SACHET (CAP)
FLOOD	GloFAS
LANDSLIDE . HISTORY	NASA GLC 2007–17
TRAFFIC	fleet GPS fixes
EVIDENCE	ROUTE-SPECIFIC EVIDENCE · FastAPI backend every factor sampled along the selected road · PostgreSQL + PostGIS (Supabase) · OSRM routes, alternatives, turn steps
POLICY	DETERMINISTIC SAFETY POLICY 11 factors · reason codes · UNKNOWN ≠ SAFE · reviewer authorisation when hazard data is missing
DECISION	CONTINUE CAUTION HOLD REROUTE
LOOP	MANAGER REVIEW DRIVER NAVIGATION ROUTE-AHEAD 60 s CONDITIONS CHANGE? REASSESS
CLIENTS	Manager console — React + TypeScript, manager accounts only · Driver app — React Native / Expo, Android; one login, server-decided role, offline trip package
STACK	FastAPI · PostgreSQL / PostGIS (Supabase) · React · React Native / Expo · OSRM · OpenStreetMap · Gemini

AUTHORITY / WHO DECIDES

DETERMINISTIC SAFETY ENGINE
makes every operational decision

GEMINI / OPENROUTER (LLM)
explains the decision in the driver's language — never decides

EXPERIMENTAL ML
research channel only; held outside routing by the safety gate

GOVERNANCE CHAIN



RUN LOG / PHYSICAL ANDROID / HOSTED BACKEND

Not a concept. A working end-to-end prototype.

PLAN

REVIEW

DISPATCH

ACCEPT

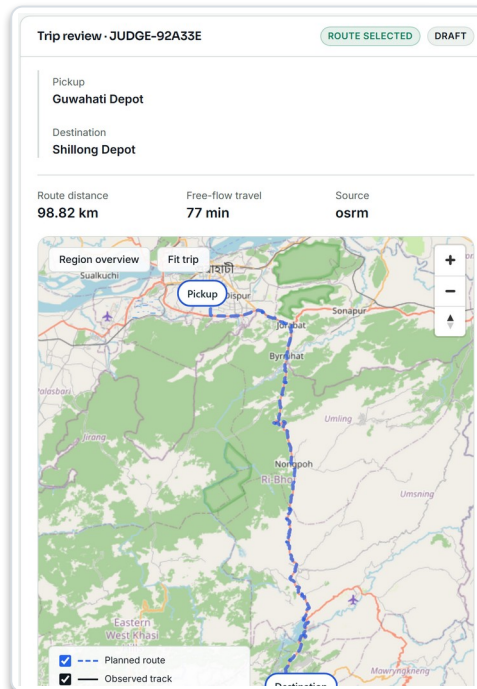
TRUCK VERIFY

NAVIGATE

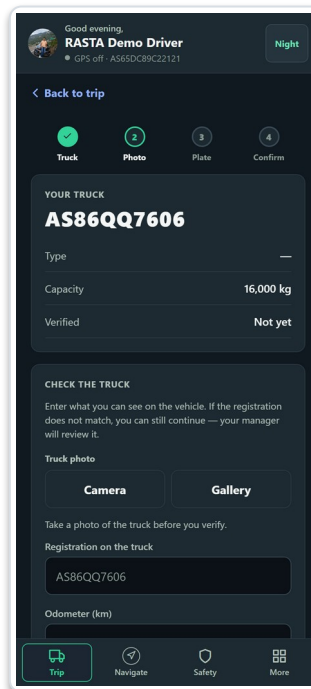
OFF-ROUTE

MANAGER REROUTE APPROVAL

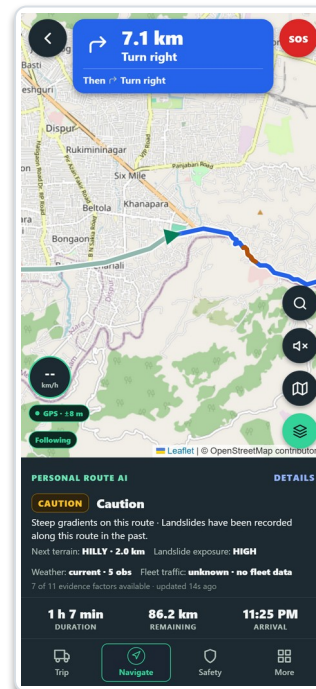
DELIVERY



MANAGER / ROUTE REVIEW



DRIVER / TRUCK CHECK



DRIVER / NAVIGATION

EVIDENCE OF EXECUTION

The whole lifecycle above was run end to end on a physical Android phone against the hosted backend, on the real road.

12/12 physical judge-flow steps

9/9 physical role-flow steps

1138+ backend · 621+ driver · 170+ manager tests

PHYSICAL ANDROID · CERTIFIED

CANONICAL DEMO: GUWAHATI → SHILLONG / ≈ 98.8 KM REAL ROUTE

KNOWN LIMITS

Render cold start → warm /health before the demo · provider rate limits → cached evidence + explicit UNKNOWN · background push needs Firebase config → in-app alerts already work · experimental ML → outside routing until the gate passes

IMPACT AND BENEFITS

IMPACT / WHO GAINS WHAT

RASTAAI does not ask only “Which road is shortest?” It asks “Can this truck reliably use this corridor now?”

FOOD · MEDICINE · FUEL · RELIEF SUPPLIES — essential logistics for hill communities, decided on evidence

FLEET . MANAGERS	Route usability and uncertainty visible before dispatch; auditable decisions.
DRIVERS	Safer navigation, danger context, emergency support, 23 languages.
REMOTE . COMMUNITIES	More resilient access to food, medicine, fuel and relief supplies.
GOVERNMENT/OPERATIONS	Auditable route evidence and disruption visibility per corridor.

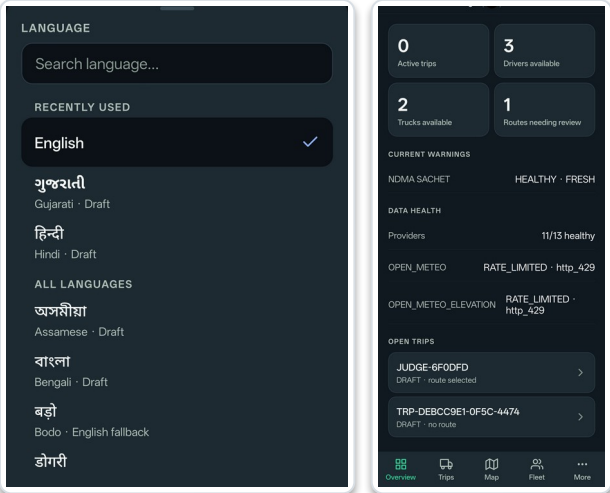
DEMONSTRATED TODAY

23	selectable languages, explicit fallback status	12/12	judge-flow steps certified on a physical phone
11	route factors in every decision	100 %	of reroutes reviewed by a human
60 s	route-ahead monitor while driving	0	unknown factors counted as safe

HUMAN GOVERNANCE — AI HAS NO UNCONTROLLED AUTHORITY



EXHIBIT C — REAL SCREENS / APK 1.0.18



DRIVER / 23 LANGUAGES

MANAGER / MOBILE

Real screens, APK 1.0.18. One login, server-decided role: a manager gets a mobile fleet view, a driver gets navigation. Each language shows its status — Verified, Draft or English fallback — so nobody is misled.

SOURCE REGISTER / EVIDENCE BEHIND THE ROUTE DECISION

ROUTING/PLACES

OSRM · OpenStreetMap · Nominatim

real road geometry, alternatives and turn steps; place search on the corridor

WEATHER/TERRAIN

Open-Meteo · MET Norway · Copernicus DEM · OpenTopoData

forecast sampled along the road; climb, gradient and steep segments (GLO-90)

OFFICIAL/FLOOD

NDMA SACHET (CAP) · Copernicus GloFAS

official warnings polled with freshness; river discharge against its 30-day mean

LANDSLIDE . HISTORY

NASA Global Landslide Cataloghistorical exposure; inventory 2007–2017, ≤ 5 km location-accuracy filter

AI . ASSISTANCE

Google Gemini

explains the decision in the driver's language — never safety authority

PROBLEM . STATEMENT

MDoNER · SIH26002

AI-based smart logistics and accessibility intelligence for the NER

REFERENCES

gpm.nasa.gov/landslides (NASA GLC) · sachet.ndma.gov.in · global-flood.emergency.copernicus.eu (GloFAS) · open-meteo.com · api.met.no · opentopodata.org · Copernicus DEM · project-osrm.org · openstreetmap.org · github.com/nxtlucifer/ner-ai-logistics (repository, tests, model registry)

EXPERIMENTAL LANDSLIDE RESEARCH

95 %

RECALL

on the NER geographic holdout — 154 catalogued events, no NER row used in training

SAFETY GATE

NOT DEPLOYED

FPR 50 % too high for production deployment

TRAIN

HELD-OUT NER TEST

USEFUL SIGNAL FOUND

FALSE POSITIVES TOO HIGH

SAFETY GATE

NOT DEPLOYED

VALIDATION BEFORE AUTOMATION. Strong recall was not enough. Held-out testing exposed the false-positive rate, so the gate keeps the model out of routing.

Precision 0.32 · F1 0.48 · PR-AUC 0.55 · ROC-AUC 0.83 · Brier 0.12 · calibrated logistic regression on NASA GLC + ERA5-Land rain + DEM slope · docs/HAZARD_ERROR_ANALYSIS.md